

A Harmonic Self-Control Network for Compact High-Efficiency Rectifiers With Extended Bandwidth

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Abstract—This letter proposes and designs a novel harmonic self-control network (HSCN). The network performs impedance matching while simultaneously suppressing the high harmonics produced by nonlinear components in a rectifier. The HSCN can achieve efficient impedance matching and enhance the rectifying efficiency simultaneously. It ensures a compact design, while the inherent broadband impedance matching capability of the HSCN expands the operational bandwidth. A theoretical analysis of the HSCN demonstrates the effectiveness of the design method. A rectifier relying on the proposed HSCN is designed, fabricated, and measured. It achieves a maximum efficiency of 81.7% at 2.1 GHz with an RF power of 16.5 dBm. At the same time, the rectifier achieves an efficiency greater than 70% from 1.8 to 2.85 GHz. The size of the rectifier is only $16.6 \times 8.7 \text{ mm}^2$. Higher efficiency, wider bandwidth, and a more compact size were achieved compared to conventional designs.

Index Terms—Broadband, energy harvesting, harmonic control, high efficiency, impedance match, rectifier.

I. INTRODUCTION

WITH the advancement of wireless power transfer (WPT) technology, its application range has been continuously expanding [1], [2], [3]. WPT can not only supply power to electronic devices such as sensors and Internet of Things (IoT) nodes over long distances—reducing the reliance on batteries and cables—but also be integrated with other advanced technologies for remote charging of mobile devices [4], [5]. These developments place higher demands on a critical component within a WPT system: the rectifier. Rectifiers are now required to be compact and easily integrated with other components while also achieving high efficiency [6], [7]. Ideally, they should be capable of adapting to variations in operating frequencies, output loads, and input power levels [8], [9], [10], [11].

Harmonic termination control is an effective and widely used design methodology by reshaping the voltage and current waveforms on a diode by controlling higher order harmonics generated during rectifying [12], [13], [14], [15], [16], [17]. This reshaping reduces the overlap between voltage and

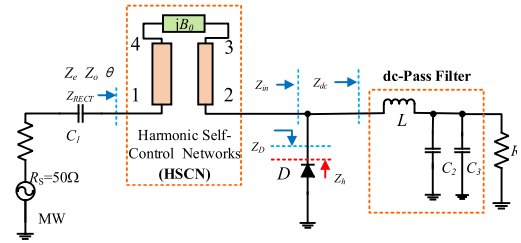


Fig. 1. Schematic of the HSCN applied to a rectifier.

current waveforms, thereby minimizing losses and improving overall power conversion efficiency (PCE). A compact Class-F harmonic termination network was proposed in [16] to achieve a short circuit at the second harmonic and an open circuit at the third harmonic, resulting in an efficiency of 80.4% at 900 MHz. In contrast, a Class-F⁻¹ harmonic termination network not only ensures high rectifying efficiency but also provides improved operational bandwidth [17]. In [18], two shorted stubs resonant at $2f_0$ and $3f_0$ were used to achieve zero impedance at harmonic frequencies.

Such designs, with their complex structural features, impose strict requirements on harmonic terminations, ultimately limiting the operational bandwidth. A continuous Class-F concept used in power amplifier design has been applied to rectifier designs, reaching over 50% efficiency with a limited bandwidth from 0.78 to 1.43 GHz [19]. To address these limitations, an improved Class-R harmonic control network sets the harmonic impedance to purely reactance. This approach significantly enhances the operational bandwidth while maintaining high rectification efficiency, as demonstrated in [20], [21], and [22].

In this work, a harmonic self-control network (HSCN) is proposed and utilized for the design of compact and efficient rectifiers with extended bandwidth. The compactness of the rectifier is achieved through a novel coupling-matching structure, which not only enables effective impedance matching but also provides standard Class-R harmonic termination control. It ensures that the rectifier maintains high PCE over a specified bandwidth. Furthermore, the overall size of the design is significantly smaller compared to conventional designs reported in the literature.

II. PRINCIPLE AND DESIGN

Fig. 1 illustrates the operating principle of HSCN applied to a rectifier design. The rectifier features a simple and compact

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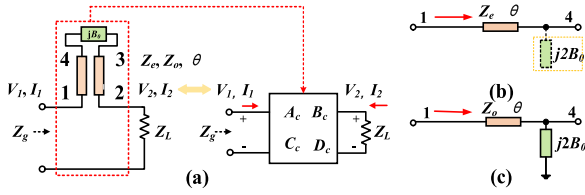


Fig. 2. (a) Schematic of the proposed HSCN and its equivalent network. (b) Even mode. (c) Odd mode.

structure, utilizing only an HSCN that simultaneously realizes input impedance matching and harmonic termination control.

A. Impedance Match Analysis

The HSCN is shown in Fig. 2, where ports 3 and 4 of the coupled lines are connected via reactive components. The electrical length of the coupled line is θ . The characteristic impedances of even and odd modes are Z_e and Z_o , respectively. Under this configuration, the four-port network can be equivalently represented as a two-port network. The even- and odd-mode analysis method is employed for the analysis, and the corresponding $ABCD$ matrices for the odd and even modes are given as follows [23]:

$$\begin{bmatrix} A_e & B_e \\ C_e & D_e \end{bmatrix} = \begin{bmatrix} \cos \theta & jZ_e \sin \theta \\ j \sin \theta / Z_e & \cos \theta \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} A_o & B_o \\ C_o & D_o \end{bmatrix} = \begin{bmatrix} \cos \theta - 2B_0 Z_o \sin \theta & jZ_o \sin \theta \\ j(\sin \theta / Z_o + 2B_0 \cos \theta) & \cos \theta \end{bmatrix} \quad (2)$$

As the even- and odd-mode analyses are applied, the voltage-current equations of the network are expressed as

$$\begin{bmatrix} \frac{1}{2}(V_1 + V_2) \\ \frac{1}{2}(I_1 + I_2) \end{bmatrix} = \begin{bmatrix} A_e & B_e \\ C_e & D_e \end{bmatrix} \begin{bmatrix} \frac{1}{2}(V_4 + V_3) \\ -\frac{1}{2}(I_4 + I_3) \end{bmatrix} \quad (3)$$

$$\begin{bmatrix} \frac{1}{2}(V_1 - V_2) \\ \frac{1}{2}(I_1 - I_2) \end{bmatrix} = \begin{bmatrix} A_o & B_o \\ C_o & D_o \end{bmatrix} \begin{bmatrix} \frac{1}{2}(V_4 - V_3) \\ -\frac{1}{2}(I_4 - I_3) \end{bmatrix} \quad (4)$$

As shown in Fig. 2, ports 3 and 4 of the HSCN are open circuit. Thus, the currents at the two ports are satisfied

$$I_3 = I_4 = 0. \quad (5)$$

Applying (5) into (3) and (4) and removing V_3 and V_4 , we obtain the equations that contain only V_1 , I_1 , V_2 , and I_2 as

$$\begin{aligned} C_e(V_1 + V_2) &= A_e(I_1 + I_2) \\ C_o(V_1 - V_2) &= A_o(I_1 - I_2). \end{aligned} \quad (6)$$

Thus, the $ABCD$ matrix expression of the coupling-matching network in Fig. 2 is derived

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \quad (7)$$

where

$$\begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} = \frac{1}{A_e C_o - A_o C_e} \begin{bmatrix} A_e C_o + A_o C_e & 2A_e A_o \\ 2C_e C_o & A_e C_o + A_o C_e \end{bmatrix}. \quad (8)$$

The HSCN directly connects the source impedance with the diode impedance, transforming the diode impedance and matching it to the source impedance. The reflection coefficient

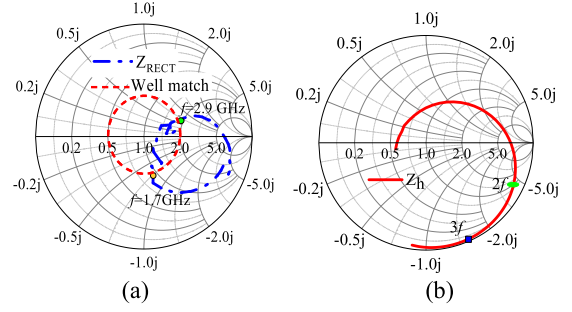


Fig. 3. (a) Broadband impedance matching effects of the HSCN. (b) Harmonic impedance variation of HSCN.

of the rectifier provides an intuitive indication of the matching condition. From (4), it is obtained that

$$\frac{A_c Z_L + B_c}{C_c Z_L + D_c} = Z_g \quad (9)$$

where Z_L is the load impedance and $Z_L = R_L + jX_L$. For an ideal matching network, achieving $S_{11} = 0$ is the design goal. Thus, when $Z_g = R_S$, $R_L \neq R_S$, (9) is simplified as

$$Z_e = \left(-M_1 + \sqrt{M_1^2 + M_2} \right) \tan \theta \quad (10)$$

$$B_0 = \frac{M_2 \tan^2 \theta + Z_e Z_o}{2(Z_e Z_o - M_2) Z_o \tan \theta} \quad (11)$$

where

$$M_1 = \frac{R_S X_L}{R_L - R_S} \quad M_2 = \frac{R_S X_L^2}{R_L - R_S} + R_S R_L. \quad (12)$$

Usually, the source impedance R_S and the load $Z_L = R_L + jX_L$ are not identical. Therefore, (10) is certainly satisfied under this condition. It ensures that the designed Z_e is a valid value and physically realizable

$$M_1^2 + M_2 > 0. \quad (13)$$

Through the above analysis, it is demonstrated that the HSCN effectively transforms any load to achieve impedance matching to the source. Furthermore, (10)–(13) indicate that Z_o and θ serve as independent variables, unaffected by either the load impedance Z_L or the source impedance R_S . These parameters may be freely adjusted, offering another degree of design flexibility for the rectifier in terms of frequency bandwidth and input power range. The broadband matching effects of the HSCN are shown in Fig. 3, where the red circle indicates a good match.

B. Class-R Harmonics Control

Through the analysis of the diode impedance, it is found that the reactive element in the HSCN is an inductor. The inductive element facilitates a good match of the rectifier within the operating frequency range. However, as the frequency increases, the inductive reactance also increases, and at higher order harmonics (e.g., second and third harmonics), the inductor behaves approximately as an open circuit. Under this condition, HSCN can be equivalently represented as a two-port network at harmonic frequencies, as illustrated in Fig. 4.

The corresponding $ABCD$ matrix at harmonic frequencies is expressed as follows:

$$\begin{bmatrix} A_h & B_h \\ C_h & D_h \end{bmatrix} = \begin{bmatrix} \frac{Z_e + Z_o}{Z_e - Z_o} & \frac{2jZ_e Z_o}{Z_e - Z_o} \\ \frac{Z_e + Z_o}{Z_e - Z_o} & \frac{Z_e + Z_o}{Z_e - Z_o} \end{bmatrix}. \quad (14)$$

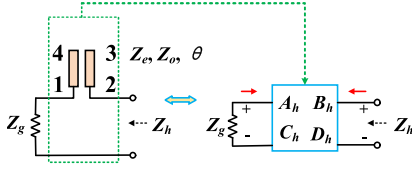


Fig. 4. Higher harmonic equivalence network of HSCN.

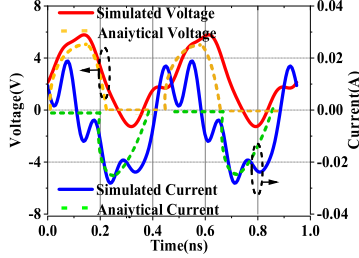


Fig. 5. Diode voltage and current operating waveforms.

Harmonic termination control primarily aims to regulate the input impedance Z_h of the diode at higher order harmonics. Based on (14), the expression for calculating Z_h is

$$Z_h = \frac{A_h Z_g + B_h}{C_h Z_g + D_h} = \frac{Z_g}{Z_g + 1} + \frac{2jZ_e Z_o \cot \theta}{(Z_g + 1)(Z_e + Z_o)} \quad (15)$$

where Z_g is the source impedance, fixed at 50Ω . From (15), we observe that the HSCN provides an approximate pure reactive impedance (with a nonzero imaginary part) to the diode, as shown in Fig. 3(b). It exhibits a characteristic of total reflection for incident power. This form of harmonic termination is referred to as Class-R harmonic control. Similar to Class-F, it improves rectification efficiency by shaping the voltage and current waveforms of the diode, as illustrated in Fig. 5. In addition, the HSCN offers relaxed harmonic termination conditions, which further broaden the effective harmonic control bandwidth.

C. Design of the Rectifier

As discussed previously, the HSCN demonstrates highly efficient impedance transformation, which contributes to an extended operational bandwidth. In addition, it effectively suppresses higher order harmonics generated by the diode, reshaping the voltage and current waveforms at its input.

The operational bandwidth of the rectifier has also been significantly improved. The center frequency of the rectifier was set near $f_0 = 2.1$ GHz, and its extended bandwidth successfully covers both $f_1 = 1.8$ GHz and $f_2 = 2.45$ GHz, which are within the IMS frequency bands. Substituting the input impedances $Z_{in}(f_1)$ and $Z_{in}(f_2)$ into (10), solve jointly to obtain θ_0 ; subsequently, substituting $Z_{in}(f_0)$ into (10) yields Z_e . For an identical inductor, employing (11) establishes $B_0(f_1)/B_0(f_2) = f_2/f_1$, enabling the determination of Z_o . Finally, B_0 is calculated. A comprehensive simulation optimization of the four variables required for the HSCN is performed, thereby achieving broadband matching for the HSCN.

The diode's operating voltage and current at 2.1 GHz with 13-dBm input power were analyzed, and the corresponding waveforms are shown in Fig. 5. It can be observed that the voltage and current waveforms alternate in time with reduced overlap, which effectively minimizes power loss and improves rectification efficiency.

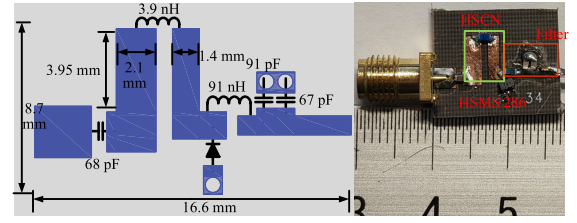


Fig. 6. Design layout and physical drawing of the rectifier.

TABLE I
SPICE PARAMETERS OF AN HSMS286 DIODE

B_v	C_{j0}	I_s	R_s	V_{bi}
7.0 V	0.18 pF	0.05 μ A	6.0 Ω	0.25 V

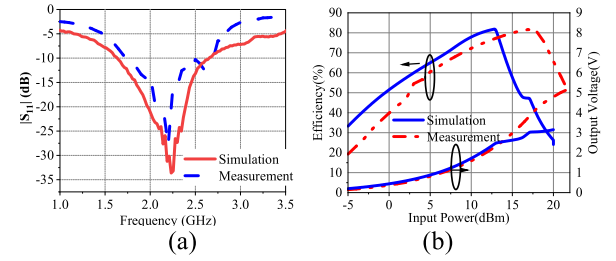


Fig. 7. (a) $|S_{11}|$ of the rectifier at 12 dBm. (b) Efficiency and output voltage of the rectifier at different inputs.

III. MEASUREMENT AND ANALYSIS

The proposed rectifier, designed based on the theory, was fabricated and experimentally evaluated. The layout and detailed design parameters of the rectifier are shown in Fig. 6, which uses an HSMS 286 diode, and its SPICE parameters are shown in Table I. The overall size of the rectifier is extremely compact, occupying an area of only 16.6×8.7 mm. The design utilizes an F4B-2 substrate with a relative dielectric constant of 2.65 and a thickness of 1 mm. The inductance used in HSCN is 3.9 nH.

The input impedance matching performance of the rectifier was measured using a vector network analyzer (VNA), and the results are shown in Fig. 7(a). Under a 12-dBm RF input, the rectifier achieves excellent impedance matching at 2.1 GHz. Additionally, its operating bandwidth is significantly extended; the S -parameter remains below -10 dB over a wide frequency range from 1.65 to 2.65 GHz, indicating good matching across the band.

The rectifier's PCE was also measured at 2.1 GHz, as shown in Fig. 7(b). At an input power of 16.5 dBm and with a 350- Ω load, the rectifier achieved a peak efficiency of 81.7%. Over a wide input power range from 2 to 22 dBm, the efficiency remains above 50%, demonstrating robust performance. Compared to simulation results, the measured peak efficiency is consistent, although the optimal input power level is slightly higher. This discrepancy is primarily attributed to the diode used in the prototype, which exhibits a slightly higher reverse breakdown voltage, leading to an increase in the optimal operating voltage.

The rectifier's performance under different dc load conditions was also characterized, as illustrated in Fig. 8(a). A frequency sweep analysis was performed on the rectifier, and the results are shown in Fig. 8(b). Under an RF input power of 16.5 dBm, the rectifier maintains an efficiency

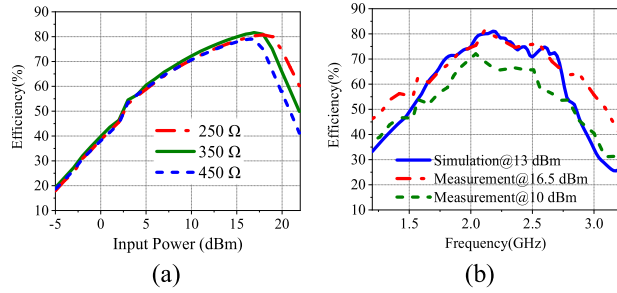


Fig. 8. (a) PCE versus frequency with different dc loads. (b) Efficiency of the rectifier at different frequencies.

TABLE II
COMPARISON WITH PREVIOUS RECTIFIERS

Ref	[12]	[15]	[17]	[22]	This Work
Frequency (GHz)	2.45	2.45	2.35	2.4	2.1
Harmonic termination	Class F	Bandstop	Class F ⁻¹	Class-R	Class-R
Peak PCE	79%	80.9%	80.4%	82.5	81.7%
Power	27 dBm	20 dBm	14 dBm	12	16.5dBm
BW (GHz)	1.85-2.6*	2-2.65*	2.3-2.45 & 2.65-2.85*	1.7-2.8	1.8-2.85
PCE >70%	33.7%	28.0%		48.9%	45.2%
BW(GHz)	1.55-2.75	1.8-2.9	NA	1.8-3.0*	1.5-3.1
PCE >50%	55.8%	46.8%		50%	69.6%
Size(mm ²)	35×30	18×16	30×19	46×32	16.6×8.7

*Graphically estimated; NA: Not Available.

above 70% across the frequency range of 1.85–2.85 GHz and above 50% from 1.5 to 3.1 GHz. These results demonstrate that the proposed rectifier achieves a significantly enhanced operational bandwidth.

Table II displays a comparative analysis of the rectifier's performance compared to previous studies. The findings reveal that the proposed rectifier achieves the highest rectification efficiency while greatly broadening the operating band for harmonic termination control. What is more surprising is the compact size of the rectifier, which is no more than half the size of other rectifiers.

IV. CONCLUSION

In this letter, an efficient and compact matching network for harmonic self-control is proposed and verified. Effective matching and harmonic termination control ensure high efficiency and a wide bandwidth of the circuit within a small design size. The validity of the theory is verified by processing and testing a rectifier. The design methodology allows for better integration with the device while improving the operating performance of the rectifier. This can be used for wireless energy transmission and harvesting in small electronic devices such as the IoT, which greatly improves the application range of the device. The proposed HSCN may be applied to nonlinear circuits, e.g., power amplifiers, rectifiers, and voltage-controlled oscillators.

REFERENCES

[1] H. He, C. Liao, Z. He, L. Yan, and C. Liu, "High-efficiency ultrawideband microwave rectifier based on adaptive harmonic control," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 12, pp. 10017–10027, Dec. 2025, doi: [10.1109/TMTT.2025.3624399](https://doi.org/10.1109/TMTT.2025.3624399).

[2] N. Shinohara, "Trends in wireless power transfer: WPT technology for energy harvesting, millimeter-wave/THz rectennas, MIMO-WPT, and advances in near-field WPT applications," *IEEE Microw. Mag.*, vol. 22, no. 1, pp. 46–59, Jan. 2021.

[3] J. Zhang, Y. Huang, and J. Zhou, "A wide dynamic range rectifier based on HEMT with a variable self-bias voltage," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 2, pp. 547–551, Feb. 2024.

[4] P. Kamalinejad, C. Mahapatra, Z. Sheng, S. Mirabbasi, V. C. M. Leung, and Y. L. Guan, "Wireless energy harvesting for the Internet of Things," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 102–108, Jun. 2015.

[5] C. Xu, Y. Zhuang, C. Song, Y. Huang, and J. Zhou, "Dynamic wireless power transfer system with an extensible charging area suitable for moving objects," *IEEE Trans. Microw. Theory Techn.*, vol. 69, no. 3, pp. 1896–1905, Mar. 2021.

[6] H. He, C. Liao, Z. He, L. Yan, and C. Liu, "An arbitrary-frequency dual-band high-efficiency rectifier employing a novel frequency-selective power allocation network," *IEEE Trans. Power Electron.*, vol. 40, no. 7, pp. 8939–8943, Jul. 2025.

[7] S. Saffari, A. Basaligheh, and K. Moez, "An RF-to-DC rectifier with high efficiency over wide input power range for RF energy harvesting applications," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 66, no. 12, pp. 4862–4875, Dec. 2019.

[8] Z. He, L. Yan, and C. Liu, "An adaptive power division strategy for nonlinear components in rectification," *IEEE Trans. Power Electron.*, vol. 39, no. 12, pp. 15436–15440, Dec. 2024.

[9] H. He, H. Lin, P. Wu, Q. Li, and C. Liu, "Compact high-efficiency broadband rectifier based on coupled transmission line," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 11, pp. 4404–4408, Nov. 2022.

[10] H. He, Y. Zhou, Z. He, Y. Feng, and C. Liu, "High-efficiency octave bandwidth rectifier for electromagnetic energy harvesting," *IEEE Microw. Wireless Technol. Lett.*, vol. 35, no. 5, pp. 549–552, May 2025.

[11] F. Cheng, C.-H. Du, L. Wu, and C. Gu, "Compact and ultra-wideband high-efficiency rectifier using asymmetric coupled-line impedance transformer," *Intern. J. Micro. Wireless Tech.*, vol. 16, no. 6, pp. 966–971, Apr. 2024.

[12] G. Moloudian, J. L. Buckley, and B. O'Flynn, "A novel rectenna with class-F harmonic structure for wireless power transfer," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 2, pp. 617–621, Feb. 2024.

[13] Z. He and C. Liu, "A compact high-efficiency broadband rectifier with a wide dynamic range of input power for energy harvesting," *IEEE Microw. Wireless Compon. Lett.*, vol. 30, no. 4, pp. 433–436, Apr. 2020.

[14] J. Shin and J. Oh, "High efficiency class-F rectifier based on harmonic-controlled voltage doubler with extended input power and frequency range," *IEEE J. Microw.*, vol. 5, no. 1, pp. 130–136, Jan. 2025.

[15] C. Liu, F. Tan, H. Zhang, and Q. He, "A novel single-diode microwave rectifier with a series band-stop structure," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 2, pp. 600–606, Feb. 2017.

[16] J. Guo, H. Zhang, and X. Zhu, "Theoretical analysis of RF-DC conversion efficiency for class-F rectifiers," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 4, pp. 977–985, Apr. 2014.

[17] F. Zhao, D. Inserra, G. Wen, J. Li, and Y. Huang, "A high-efficiency inverse class-F microwave rectifier for wireless power transmission," *IEEE Microw. Wireless Compon. Lett.*, vol. 29, no. 11, pp. 725–728, Nov. 2019.

[18] M. Roberg, T. Reveyrand, I. Ramos, E. A. Falkenstein, and Z. Popovic, "High-efficiency harmonically terminated diode and transistor rectifiers," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 12, pp. 4043–4052, Dec. 2012.

[19] H. Zhang and X. Zhu, "A broadband high efficiency rectifier for ambient RF energy harvesting," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Tampa, FL, USA, Jun. 2014, pp. 1–3.

[20] K. Miwatashi and N. Shinohara, "Development of class-R rectifier for microwave wireless power transmission to EV trucks," in *Proc. IEEE Wireless Power Transf. Conf. (WPTC)*, San Diego, CA, USA, Jun. 2021, pp. 1–4.

[21] D.-A. Nguyen and C. Seo, "Design of high-efficiency broadband rectifier with harmonic control for wireless power transfer and energy harvesting," *IEEE Microw. Wireless Compon. Lett.*, vol. 32, no. 10, pp. 1231–1234, Oct. 2022.

[22] D.-A. Nguyen, K. Woo, G. T. Bui, H. Nam, and C. Seo, "Wide dynamic-range and high-efficiency shunt-diode rectifier based on class-R harmonic control and resistance compression network," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 4, pp. 2359–2363, Apr. 2024.

[23] D. M. Pozar and R. E. M. Collin, *Foundations for Microwave Engineering*, 2nd ed., New York, NY, USA: McGraw-Hill, 1992.